Lecture Slides: Division

• SLIDE 1: Division**

• Subtitle:* Sharing things equally and counting groups

• What division means

• How to find the answer

• What a remainder is

• Speaker note:* Introduce the idea that division is a way to split a whole into equal parts or to count how many groups we can make.

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• SLIDE 2: Learning Objectives**

By the end of this lesson you will be able to:

• Name the **dividend**, **divisor**, **quotient**, and **remainder**

• Write a division sentence for a sharing or grouping situation

• Solve a division problem in **two** different ways

• Explain when a remainder appears

• Speaker note:* Read the objectives aloud and ask students to give a quick example of each term.

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• SLIDE 3: Why We Use Division**

• We split a pile into equal parts (e.g., sharing snacks)

• It is the opposite of multiplication

• We use it every day - packing, sharing money, telling time

• The symbols **÷** or **/** show a division problem

• Speaker note:* Tell a short story about sharing a pizza with friends and write the corresponding division sentence on the board.

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• SLIDE 4: The Four Words of Division**

• **Dividend** - the whole amount we start with

• **Divisor** - how many equal groups we want

• **Quotient** - how many items are in each group

• **Remainder** - what is left over when the groups are not exact

• Speaker note:* Use 12 counters, split them into 4 groups, and label each part with the four words.

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• SLIDE 5: Two Ways to Think About Division**

• **Sharing** - give the same number of objects to each friend

• **Grouping** - make a certain number of groups and count the objects in one group

• Example:* **12 ÷ 3 = 4** means

• three groups with four objects each, **or**

• three friends each get four objects

• Speaker note:* Show two pictures (one of sharing, one of grouping) and ask the class which picture feels easier to understand.

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• SLIDE 6: Simple Strategies to Solve Division**

• **Equal-sharing** - use real objects or drawings to make equal groups

• **Repeated subtraction** - keep taking away the divisor and count how many times you do it

• **Array** - draw rows and columns; each row is a group

• **Number-line jumps** - hop forward by the divisor until you reach the dividend

• Speaker note:* Demonstrate one problem with counters, then solve the same problem on paper using a different method.

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• SLIDE 7: Exact Division and Remainders**

• If nothing is left over, the division is **exact** (remainder = 0)

• A remainder is always **smaller** than the divisor

• Example with remainder: **17 ÷ 5 = 3 R 2** (three groups of five, two left over)

• **Fact families** connect multiplication and division:

$4 \times 3 = 12 \rightarrow 12 \div 4 = 3$ and $12 \div 3 = 4$

• Speaker note:* Show a fact-family card, read the three related sentences aloud, and ask students to create their own fact family using a number they choose.

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• SLIDE 8: Real-World Examples**

• 24 cookies, 6 per bag → **24 ÷ 6 = 4** bags

• 18 stickers, 5 students → each gets **3** stickers, **3** stickers left over

• 30 apples, crates hold 7 → **4** full crates, **2** apples left over

• $50 shared by 8 friends → each gets **$6**, **$2** left over

• 125 seconds → **2** minutes and **5** seconds (125 ÷ 60 = 2 R 5)

• Speaker note:* Invite a student to share another everyday situation that needs division and write the sentence on the board.

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• SLIDE 9: Thinking About Division**

• Why must the remainder be smaller than the divisor?

• Which strategy feels fastest for a given problem?

• What happens if we swap the dividend and divisor?

• How can fact families help you spot a mistake?

• How does division link to multiplication and subtraction?

• Speaker note:* Facilitate a brief "think-pair-share" where students discuss these questions with a partner, then share a few ideas with the whole class.

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• SLIDE 10: Summary**

• Division = sharing a whole into equal parts **or** counting groups

• Key words: dividend, divisor, quotient, remainder

• Methods: sharing, repeated subtraction, arrays, number-line jumps

• Fact families help you check your work

• Discussion prompt:* "If you could make a game that uses division, what would the rules be?"

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